3 **(a)** Any one advantage.

- No break in the communication link.
- High altitude so no significant atmosphere and hence no power require to overcome air resistance.

Any one disadvantage.

- High altitude so it requires a larger amount of fuel to get into orbit than lower orbits.
- High altitude so there is a significant lag time in the communication signal.

(b) **(i)**

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{24 \times (60 \times 60)}$$

$$= 7.272 \times 10^{-5}$$

$$(= 7.3 \times 10^{-5})$$

(ii) Gravitational force provides the centripetal force on the satellite.

$$\left(mr\omega^2 = \frac{GMm}{r^2} \right)$$

$$r\omega^2 = \frac{GM}{r^2}$$

$$r^3 = \frac{(6.67 \times 10^{-11})(5.9 \times 10^{24})}{(7.3 \times 10^{-5})^2}$$

$$r = 4.2 \times 10^7 \text{ m}$$

(c) **(i)**

$$v = r\omega$$

$$= (4.2 \times 10^7)(7.3 \times 10^{-5})$$

$$(= 3066 \text{ m s}^{-1})$$

$$KE = \frac{1}{2}mv^2$$

$$KE = \frac{1}{2}(5.9 \times 10^3)(3066)^2$$

$$KE = 2.76 \times 10^{10} \text{ J}$$

(II)

$$\Delta U = U_{final} - U_{initial} = \left(-\frac{GMm}{r_{final}} \right) - \left(-\frac{GMm}{r_{initial}} \right)$$

$$= (6.67 \times 10^{-11})(5.9 \times 10^{24})(5.9 \times 10^3) \left[\left(-\frac{1}{4.2 \times 10^7} \right) - \left(-\frac{1}{6.4 \times 10^6} \right) \right]$$

$$(= 3.075 \times 10^{11} \text{ J})$$

$$TE = \Delta U + KE$$

$$= (3.075 \times 10^{11}) + (2.76 \times 10^{10})$$

$$= 3.35 \times 10^{11} J$$

(iii) Energy is needed to overcome air resistance.